



Syllabus: Econ PhD Code Camp

USC Department of Economics, 2026

Matt Delventhal

Class: Mon-Thu 4-6pm (Pacific Time), ([Zoom](#))

Instructor: Matt Delventhal

Email: delventh@usc.edu

Office Hours: by appointment

Class Materials: <https://github.com/mdelventhal/econphd-codecamp>

Course Description

In this class, you will have the opportunity to:

1. Learn about some basic programming concepts and tools used by Economists.
2. Reinforce your math knowledge by learning how to program hands-on, concrete examples of key concepts.
3. Build a small portfolio of personalized code examples you can save and build on later during your PhD classes and beyond.
4. Establish a solid familiarity with the popular Python language, with optional extension to other languages: Matlab, R, Julia.

Each class session will be structured to include approximately 45 minutes of lecture and 75 minutes of guided practice. Everything you need for the class will be provided through an [open-source repository](#) on GitHub. Judgment-free feedback will be offered on all work that you turn in.

Course Outline

The following is a tentative schedule of topics and exercises:

- **3 Aug:** *Intro & “Hello, World!”*
 - greetings and course overview
 - ensure Colab and any other requirements are functional
 - write first basic “Hello World” programs
- **4 Aug:** *The Basics: Variables and Loops*
 - variable types and declarations
 - loops

PRACTICAL EXERCISE: calculate some sequences and limits

- **5 Aug:** *Plotting Asymptotes & Discontinuous Functions*
 - quick intro to LaTeX for typing up formulas
 - how to construct evenly-spaced sequences of numbers
 - how to implement equations in code
 - basic plotting introduction

PRACTICAL EXERCISE: plot some discontinuous functions

- **6 Aug:** *Defining Functions and Organizing Code + Taylor Expansion Demo*

- key concept: define functions for reusable code
- key concept: organize reusable code across multiple files

PRACTICAL EXERCISE: demonstrate Taylor Expansion

- **10 Aug:** *Coding Linear Algebra*

- review of linear algebra functionality of Python
- how to type matrices in LaTeX

PRACTICAL EXERCISE: calculate determinants + eigenvectors for 2x2 matrices

- **11 Aug:** *Numerical Differentiation and Integration*

- differentiation techniques, & the connection to the proof of the Fundamental Theorem of Calculus and Taylor Expansion
- integration techniques, & their connection to the Fundamental Theorem of Calculus

PRACTICAL EXERCISE: numerically prove some counterintuitive differentiation and integration results

- **12 Aug:** *A Gentle Introduction to Object-Oriented Programming*

- conceptual overview of Object-Oriented Programming and alternative paradigms
- simple examples of class definitions

PRACTICAL EXERCISE: numerically prove L'Hopital's Rule

- **13 Aug:** *Numerical Optimization Intro: Grid Search + Measuring Performance*

- overview of grid search methods for finding solutions and optima
- overview of computational performance metrics
- introduction to 3d plotting

PRACTICAL EXERCISE: grid search optimization over 1, 2, and 3 variables

- **17 Aug:** *Numerical Optimization: More Search Algorithms*

- overview of derivative and non-derivative-based algorithms such as Newton’s Method, Nelder-Mead, and gradient descent

PRACTICAL EXERCISE: implement Newton’s Method

- **18 Aug:** *Data Input and Manipulation*

- introduction to loading and manipulating data

PRACTICAL EXERCISE: evaluate “normality” of some empirical distributions

- **19 Aug:** *Numerical Stats, I*

- overview of stat and probability functionality: random numbers, built-in density/CDF functions, etc.

PRACTICAL EXERCISE: numerically prove Law of Large Numbers

- **20 Aug:** *Numerical Stats, II*

- explore and plot a variety of standard distributions
- conceptual review of Central Limit Theorem

PRACTICAL EXERCISE: numerically prove Central Limit Theorem

Statement on Academic Conduct and Support Systems

- **Academic Conduct**

- Plagiarism – presenting someone else’s ideas as your own, either verbatim or re-cast in your own words—is a serious academic offense with serious consequences. Please familiarize yourself with the discussion of plagiarism in SCampus in Part B, Section 11, “Behavior Violating University Standards” policy.usc.edu/scampus-part-b. Other forms of academic dishonesty are equally unacceptable. See additional information in SCampus and university policies on scientific misconduct, policy.usc.edu/scientific-misconduct.

- **Support Systems**

- *Counseling and Mental Health* - (213) 740-9355 - 24/7 on call studenthealth.usc.edu/counseling. Free and confidential mental health treatment for students.
- *National Suicide Prevention Lifeline* - 1 (800) 273-8255 - 24/7 on call suicidepreventionlifeline.org. Free and confidential emotional support to people in suicidal crisis or emotional distress 24 hours a day, 7 days a week.
- *Relationship and Sexual Violence Prevention Services (RSVP)* - (213) 740-9355(WELL), press “0” after hours. 24/7 on call studenthealth.usc.edu/sexual-assault. Free and confidential therapy, workshops, and training for situations related to gender-based harm.
- *Office of Equity and Diversity (OED)* - (213) 740-5086 — Title IX - (213) 821-8298 equity.usc.edu, titleix.usc.edu. Information about how to get help or help someone affected by harassment or discrimination, rights of protected classes, reporting options, and additional resources for students, faculty, staff, visitors, and applicants.
- *Reporting Incidents of Bias or Harassment* - (213) 740-5086 or (213) 821-8298 usc-advocate.symplicity.com/care_report. Avenue to report incidents of bias, hate crimes, and microaggressions to the Office of Equity and Diversity/Title IX.
- *The Office of Disability Services and Programs* - (213) 740-0776 dsp.usc.edu. Support and accommodations for students with disabilities. Services include assistance in providing readers/notetakers/interpreters, special accommodations for

test taking needs, assistance with architectural barriers, assistive technology, and support for individual needs.

- *USC Campus Support and Intervention* - (213) 821-4710 campussupport.usc.edu. Assists students and families in resolving complex personal, financial, and academic issues.
- *Diversity at USC* - (213) 740-2101 diversity.usc.edu. Information on events, programs and training from the Provost's Diversity and Inclusion Council, Diversity.
- *USC Emergency* - *UPC*: (213) 740-4321, *HSC*: (323) 442-1000 - 24/7 on call dps.usc.edu, emergency.usc.edu. Emergency assistance and avenue to report a crime. Latest updates regarding safety, including ways in which instruction will be continued if an officially declared emergency makes travel to campus infeasible.
- *USC Department of Public Safety* - *UPC*: (213) 740-6000, *HSC*: (323) 442-120 - 24/7 on call dps.usc.edu. Non-emergency assistance or information.
- *Office of the Ombuds* - (213) 821-9556 (*UPC*) / (323-442-0382 (*HSC*) ombuds.usc.edu. A safe and confidential place to share your USC-related issues with a University Ombuds who will work with you to explore options or paths to manage your concern.